

Integral Calculation Report

ModernCalculator - Numerical Methods Procedure

Problem Statement

$$\int_0^3 \int_0^z \int_0^z x + y + z \, dx \, dy \, dz$$

LaTeX Code (For copying and embedding)

```
\int_{0}^{3}\int_{0}^{z}\int_{0}^{z} x + y + z \, dx \, dy \, dz
```

Step-by-step Resolution

Parsed function

$$x + y + z$$

Detected differentials: dx, dy, dz

Step 1: Evaluating dx

$$\begin{aligned} \int_0^z (x + y + z) dx &= \left[\frac{x^2}{2} + x(y + z) \right]_0^z \\ &= \left(\frac{z^2}{2} + z(y + z) \right) - (0) \\ &= \frac{z(2y + 3z)}{2} \end{aligned}$$

Applying the Fundamental Theorem of Calculus

Step 2: Evaluating dy

$$\begin{aligned} \int_0^z \left(\frac{z(2y + 3z)}{2} \right) dy &= \left[\frac{y^2 z}{2} + \frac{3yz^2}{2} \right]_0^z \\ &= (2z^3) - (0) \\ &= 2z^3 \end{aligned}$$

Applying the Fundamental Theorem of Calculus

Step 3: Evaluating dz

$$\begin{aligned} \int_0^3 (2z^3) dz &= \left[\frac{z^4}{2} \right]_0^3 \\ &= \left(\frac{81}{2} \right) - (0) \\ &= \frac{81}{2} \end{aligned}$$

Applying the Fundamental Theorem of Calculus

Final result

$$\text{Analytical} = \frac{81}{2}$$

Numerical approximation = 40.5 | Error margin = 5.956e-13